



Setting Up GPU Drivers on RK3588

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```
camera_engine_rkaiq_rk3562_arm64.deb
camera_engine_rkaiq_rk3568_arm64.deb
camera_engine_rkaiq_rk3588_arm64.deb
camera_engine_rkisp_v2.3.0_arm64.deb
libmali-bifrost-g31-g13p0-x11-gbm_1.9-1_arm64.deb
libmali-bifrost-g31-g13p0-x11-gbm-dbgsym_1.9-1_arm64.deb
libmali-bifrost-g52-g13p0-x11-gbm_1.9-1_arm64.deb
libmali-bifrost-g52-g13p0-x11-gbm-dbgsym_1.9-1_arm64.deb
libmali-midgard-t86x-r18p0-x11-gbm_1.9-1_arm64.deb
libmali-midgard-t86x-r18p0-x11-gbm-dbgsym_1.9-1_arm64.deb
```

```
libmali-utgard-450-r7p0-x11-gbm_1.9-1_arm64.deb  
libmali-utgard-450-r7p0-x11-gbm-dbgsym_1.9-1_arm64.deb  
libmali-valhall-g610-g13p0-x11-gbm_1.9-1_arm64.deb  
libmali-valhall-g610-g13p0-x11-gbm-dbgsym_1.9-1_arm64.deb  
rknpu2.tar
```

MLC LLM is a universal deployment solution that allows efficient CPU/GPU code generation without AutoTVM-based performance tuning. This guide focuses on setting up a generic GPU environment and troubleshooting common issues on the RK3588 (RK3588 based SBC).

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#Prerequisites

Before beginning the installation process, ensure that you have:

- An RK3588 (RK3588 based Single Board Computer)
- A stable internet connection
- Basic knowledge of using the Linux command line
- Physical access to the RK3588 for setup

#OpenCL SDK Installation

The OpenCL SDK is essential if you plan to build your own models for the OpenCL backend. Follow the steps below to install the OpenCL SDK:

1. [Access OpenCL's GitHub Repository:](#)

Visit the [OpenCL SDK GitHub Repository](#) for detailed installation instructions and resources.

#RK3588 Setup

This section guides you through setting up the RK3588 for running models using the OpenCL backend.

#Installing Ubuntu 22.04

1. Download Ubuntu 22.04 Image:

Download the Ubuntu 22.04 image tailored for the RK3588 from [here](#).

2. Flash Ubuntu to SD Card:

Use a tool like `balenaEtcher` or `Rufus` to flash the downloaded image onto an SD card.

3. Boot RK3588:

Insert the SD card into your RK3588 and power it on. Follow the on-screen instructions to complete the initial setup.

#Installing Mali GPU Driver

To enable the Mali GPU for OpenCL operations, install the Mali GPU driver.

1. Download `libmali-g610.so`:

```
cd /usr/lib
sudo wget
https://github.com/JeffyCN/mirrors/raw/libmali/lib/aarch64-
linux-gnu/libmali-valhall-g610-g6p0-x11-wayland-gbm.so -O
libmali-g610.so
```

2. Ensure Firmware File Exists:

Verify if `mali_csffw.bin` exists in the `/lib/firmware` directory:

```
cd /lib/firmware  
ls mali_csffw.bin
```

If the file does not exist, download it:

```
sudo wget  
https://github.com/JeffyCN/mirrors/raw/libmali/firmware/g610/mali\_csffw.bin
```

#Configuring OpenCL ICD Loader

Set up the OpenCL Installable Client Driver (ICD) loader to recognize the Mali GPU.

1. Update Package List:

```
sudo apt update
```

2. Install Mesa OpenCL ICD:

```
sudo apt install mesa-opencl-icd -y
```

3. Create OpenCL Vendors Directory:

```
sudo mkdir -p /etc/OpenCL/vendors
```

4. Add Mali ICD Configuration:

```
echo "/usr/lib/libmali-g610.so" | sudo tee  
/etc/OpenCL/vendors/mali.icd
```

#Installing OpenCL Libraries

Install the OpenCL development libraries required for compiling OpenCL applications.

```
sudo apt install ocl-icd-opencl-dev -y
```

#Installing Dependencies

Install necessary dependencies for the Mali OpenCL backend.

```
sudo apt install libxcb-dri2-0 libxcb-dri3-0 libwayland-client0  
libwayland-server0 libx11-xcb1 -y
```

#Verifying Installation with clinfo

1. Install clinfo:

```
sudo apt install clinfo -y
```

2. Run clinfo to Validate Installation:

```
clinfo
```

Expected Output:

Look for GPU information similar to the following in the output:

```
arm_release_ver: g13p0-01eac0, rk_so_ver: 3  
Number of platforms: 2  
Platform Name: ARM Platform  
Platform Vendor: ARM  
Platform Version: OpenCL 2.1 v1.g6p0-  
01eac0.2819f9d4dbe0b5a2f89c835d8484f9cd  
Platform Profile: FULL_PROFILE  
...
```

This output confirms that the OpenCL runtime and Mali GPU driver are correctly installed and recognized by the system.

#Troubleshooting

If you encounter issues during the installation process, consider the following steps:

Check Network Connectivity: Ensure that your RK3588 has a stable internet connection for downloading packages and dependencies.

Verify File Downloads: Re-download any files if the download was interrupted or corrupted.

Review Command Outputs: Carefully read any error messages during command execution to identify missing dependencies or permissions issues.

Consult Logs: Check system logs for detailed error information using `dmesg` or reviewing logs in `/var/log/`.

#References

1. [OpenCL SDK GitHub Repository](#)
2. [MLC LLM GitHub Repository](#)
3. [Ubuntu Rockchip Releases](#)
4. [Mali GPU Drivers](#)
5. [Orange Pi Official Website](#)
6. [clinfo Documentation](#)
7. [Ubuntu Official Documentation](#)

By following this comprehensive guide, you should be able to successfully set up GPU drivers and SDKs on your RK3588, enabling efficient OpenCL backend operations with MLC LLM.

URL: <https://ib.bsb.br/setting-up-gpu-drivers-on-rk3588>

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